

Code-Layout, Readability and Reusability

Date	15 March 2024
Team ID	
Project Name	Rising Waters
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Proper formatting maintained across Python, HTML, CSS, and JavaScript files
2	Proper File Structure	Files and folders are logically organized	Yes	Standard Flask folder structure (templates, static, model)
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Descriptive names used throughout the project
4	Function / Method Names	Functions are descriptively named	Yes	Modular functions for prediction and preprocessing
5	Code Comments	Inline and block comments explain logic	Yes	Comments added for important sections
6	Modular Design	Code is split into reusable functions/modules	Yes	Flask routes, template
7	No Redundant Code	Duplicate or unused code is removed	Yes	Reusable functions minimize duplication
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Input validation and exception handling

				implemented
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Reusable Components / Modules:

S.No	Component / Module Name	Language / Technolog	Where Reused	Reusability Level (High / Medium / Low)
1	Prediction Module	Python (Flask)	Prediction Route	High

2	Decision Tree Model	Scikit-learn	Flood Prediction	High
3	HTML Templates	HTML5, Jinja2	Home, Prediction & Result Pages	High
4	CSS & Bootstrap	CSS3, Bootstrap 5	Entire Web Application	High
5	JavaScript Functions	JavaScript	Form Validation & UI Effects	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized Flask project with standard architecture
Readability	5	Clean, readable code with meaningful naming conventions
Reusability	5	Reusable modules and templates improve maintainability
Documentation / Comments	4	Important sections are documented with comments
Overall Score	4.8 / 5	Modular, maintainable, reusable, and production-ready implementation